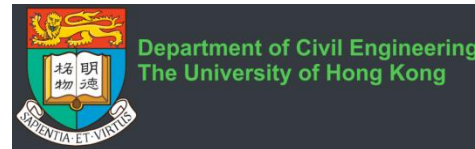


A Brief Introduction to Deep Learning and Computer Vision

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A Brief Introduction to Deep Computer Vision: Outline

✿ Basics of **Deep Learning**

- ✧ Background
- ✧ Neural Network (NN)
- ✧ Common NN architectures: Transformer

✿ Basics of **Computer Vision**

- ✧ Background
- ✧ Convolutional Neural Network (CNN)
- ✧ Common CNN architectures: ResNet

✿ Project I: **MiC Quality Assessment**

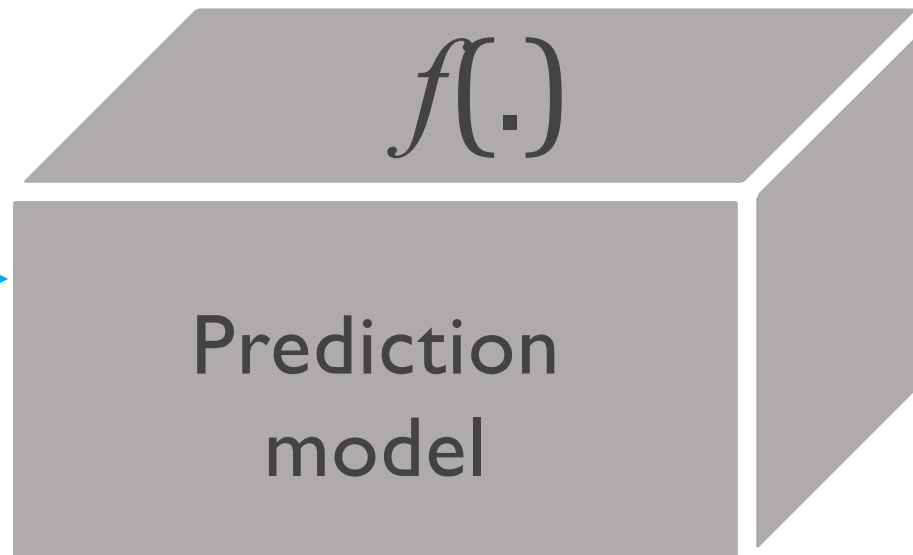
- ✧ Project description/tasks/evaluations

Basics of Deep Learning: Background

✿ Prediction Model

- ✿ Human beings
- ✿ Mathematical model: physical model

Input

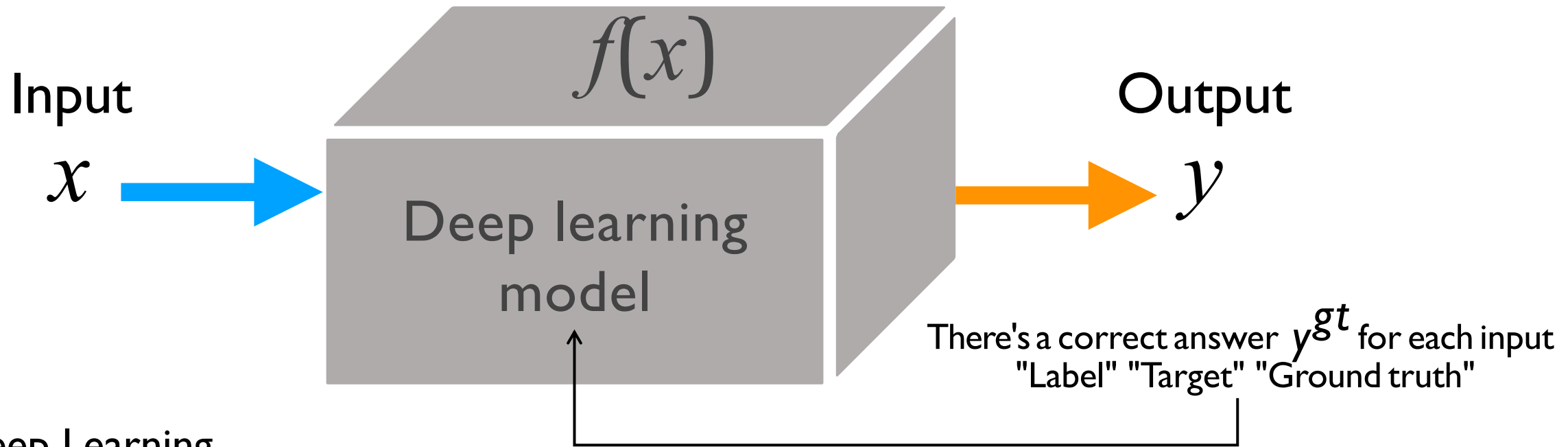


Output
“Cat”

Correct or incorrect
prediction?

[1] IMSE7140 machine learning. Qin Shujia.

Basics of Deep Learning: Background



✿ Deep Learning

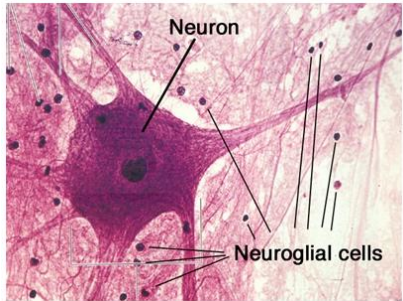
- ✿ Mathematical model: deep learning model
- ✿ Learning: learn/achieve a model that predicts input
- ✿ Deep: build the model using **neural networks (NNs)**

Utilize the correct answers to fit or train the model $f(x)$

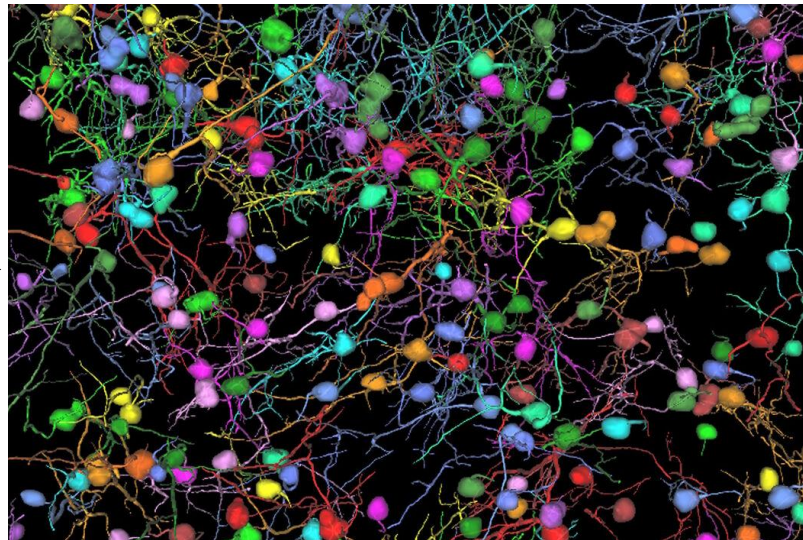
Basics of Deep Learning: Neural Network

✿ Architecture

- ✿ Human brain
- ✿ Connections of neuron cells

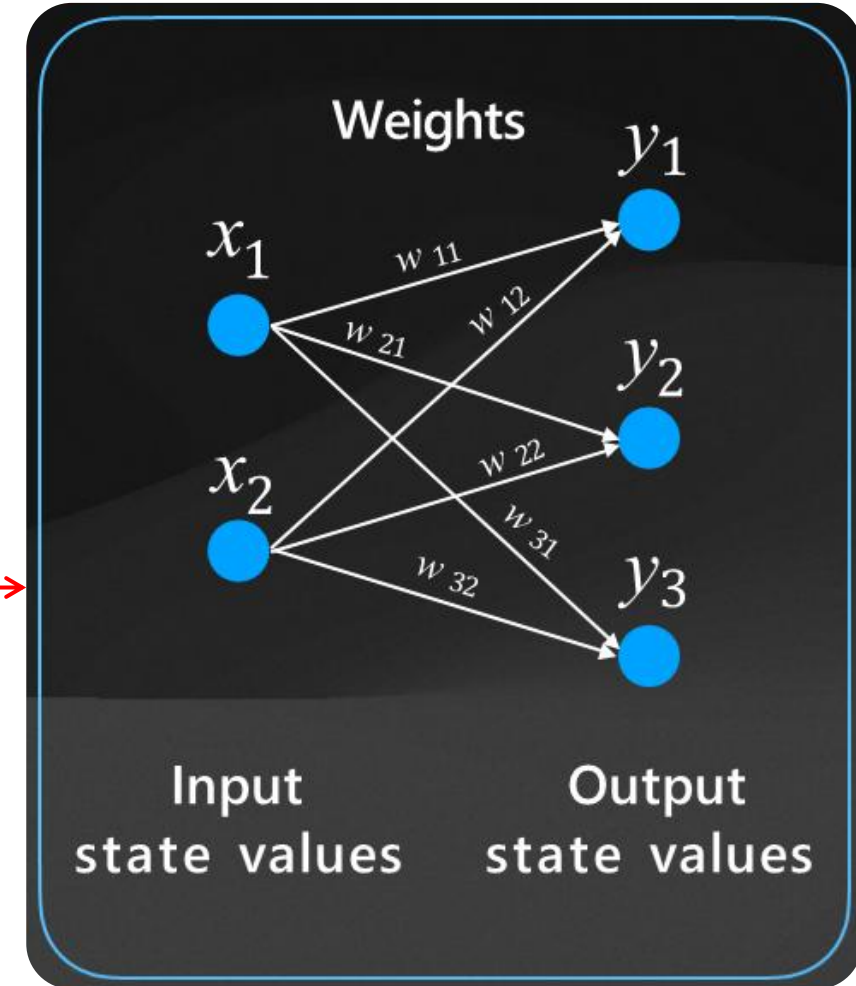


Neuron [1]



Visualization of parts of neurons [2]

Same?



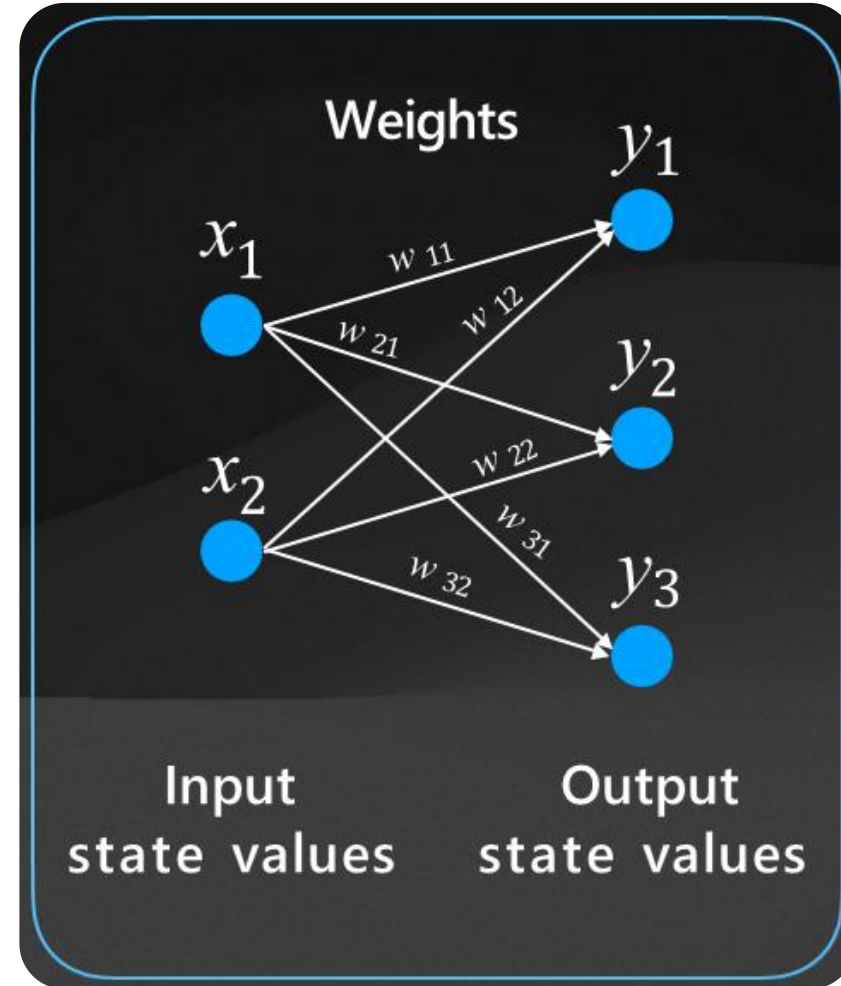
[1] <http://www.tswongsir-runners.guide/articles/neuron.htm>

[2] LEKA Microsystems.

Basics of Deep Learning: Neural Network

✿ Architecture

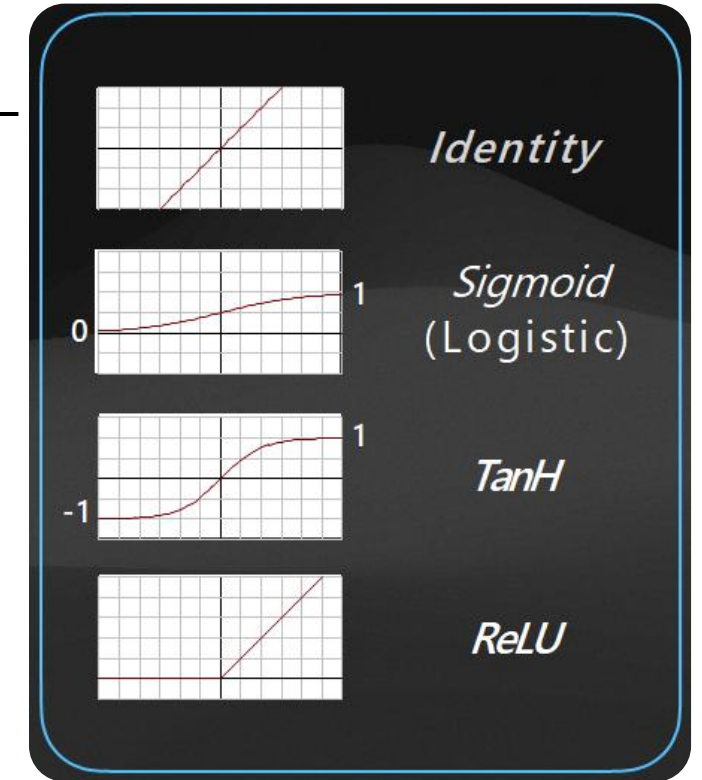
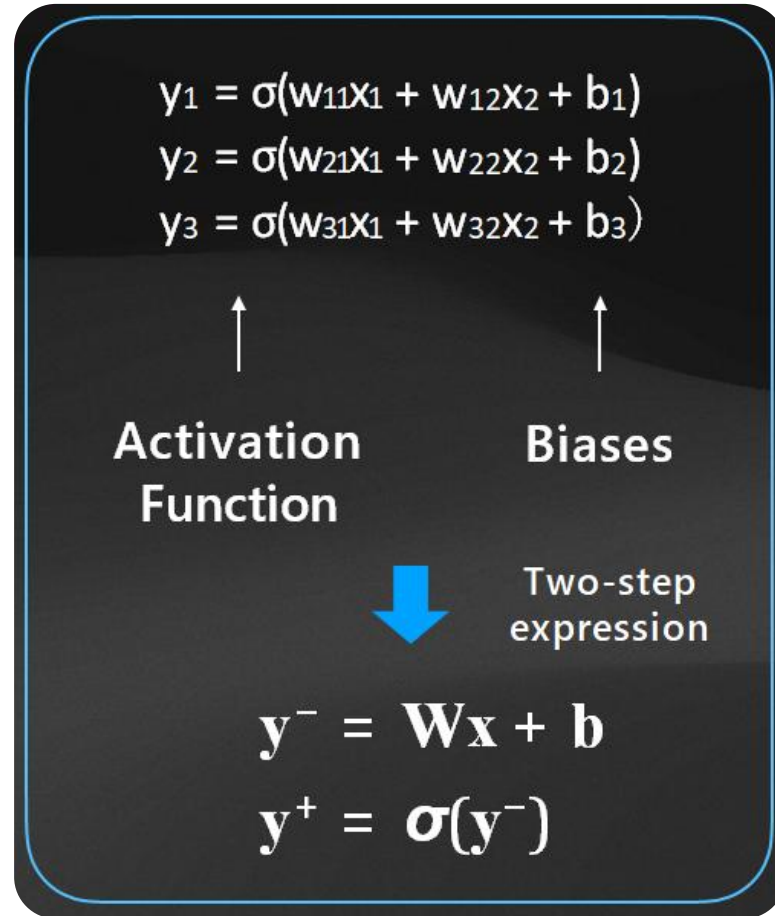
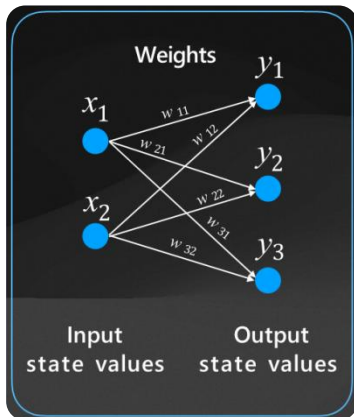
- ✿ Built by nodes/neurons in layers
- ✿ **Fully-connected (FC)** architecture
- ✿ The weights are introduced on each the connection between neurons
- ✿ The number of weights depends on the input/output dimension



Basics of Deep Learning: Neural Network

✿ Node

- ✿ Input: weighted sum
- ✿ Activation function:
learn nonlinearity



Activation functions

Basics of Deep Learning: Neural Network

✿ Optimization of NN (Backpropagation)

- ✿ Compute gradients for weights

$$\mathbf{y} = f(\mathbf{X} | \mathbf{W})$$

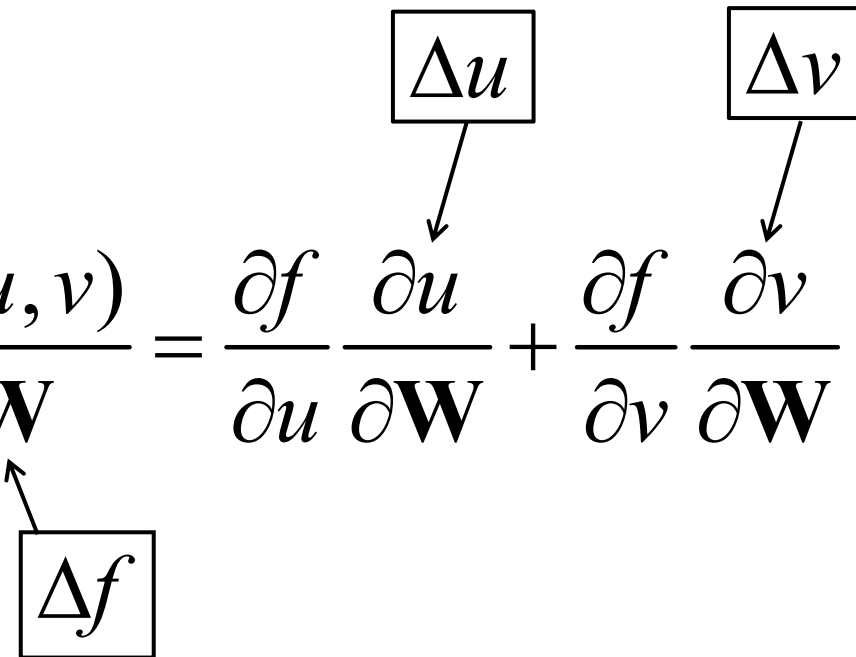
$$\mathbf{Grad} = \frac{\partial f(\mathbf{X} | \mathbf{W})}{\partial \mathbf{W}}$$

- ✿ Derivatives with respect to weights are computed
- ✿ Computations are done on the vector level

Basics of Deep Learning: Neural Network

✿ Optimization of NN (Backpropagation)

- ✿ See f as the composite function of u and v
- ✿ Change rates in directions u and v
- ✿ Total change rate: sum of change rates in all directions

$$\mathbf{y} = f(u, v)$$
$$\mathbf{Grad} = \frac{\partial f(u, v)}{\partial \mathbf{W}} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial \mathbf{W}} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial \mathbf{W}}$$


Basics of Deep Learning: Neural Network

✿ Optimization of NN (Backpropagation)

✿ Update weights using **gradient descent**

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \alpha \cdot \mathbf{Grad}$$

**Basic form of NN
optimization algorithm**

✿ Update weights using Adam optimization [1]

Require: α : Stepsize
Require: $\beta_1, \beta_2 \in [0, 1)$: Exponential decay rates for the moment estimates
Require: $f(\theta)$: Stochastic objective function with parameters θ
Require: θ_0 : Initial parameter vector

$m_0 \leftarrow 0$ (Initialize 1st moment vector)
 $v_0 \leftarrow 0$ (Initialize 2nd moment vector)
 $t \leftarrow 0$ (Initialize timestep)
while θ_t not converged **do**
 $t \leftarrow t + 1$
 $g_t \leftarrow \nabla_{\theta} f_t(\theta_{t-1})$ (Get gradients w.r.t. stochastic objective at timestep t)
 $m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (Update biased first moment estimate)
 $v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$ (Update biased second raw moment estimate)
 $\hat{m}_t \leftarrow m_t / (1 - \beta_1^t)$ (Compute bias-corrected first moment estimate)
 $\hat{v}_t \leftarrow v_t / (1 - \beta_2^t)$ (Compute bias-corrected second raw moment estimate)
 $\theta_t \leftarrow \theta_{t-1} - \alpha \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$ (Update parameters)
end while
return θ_t (Resulting parameters)

**Adam optimization algorithm
converges faster!**

[1] Kingma, D.P. & Ba, J. Adam: A method for stochastic optimization. ICLR 2015.

Basics of Deep Learning: Neural Network

✿ Optimization of NN (loss function)

- ✿ Mean square error function

- ✿ Compare prediction and ground truth using Euclidean distance

$$L = \frac{1}{NC} \sum_{i=1}^N \sum_{j=1}^C (\hat{p}_{i,j} - y_{i,j}^{gt})^2$$

Basics of Deep Learning: Neural Network

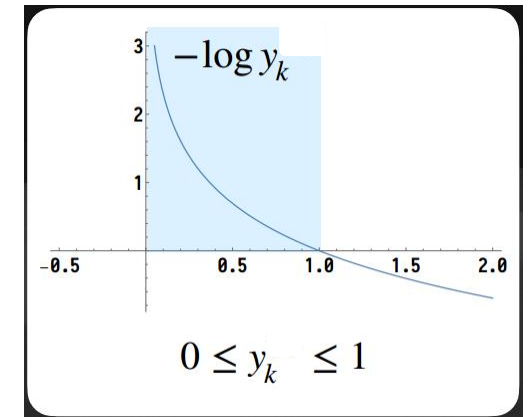
✿ Optimization of NN (loss function)

✿ Cross entropy function:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C y_{i,j}^{gt} \log(\hat{p}_{i,j})$$

True probability distribution (one-hot vector)

Model's predicted probability distribution

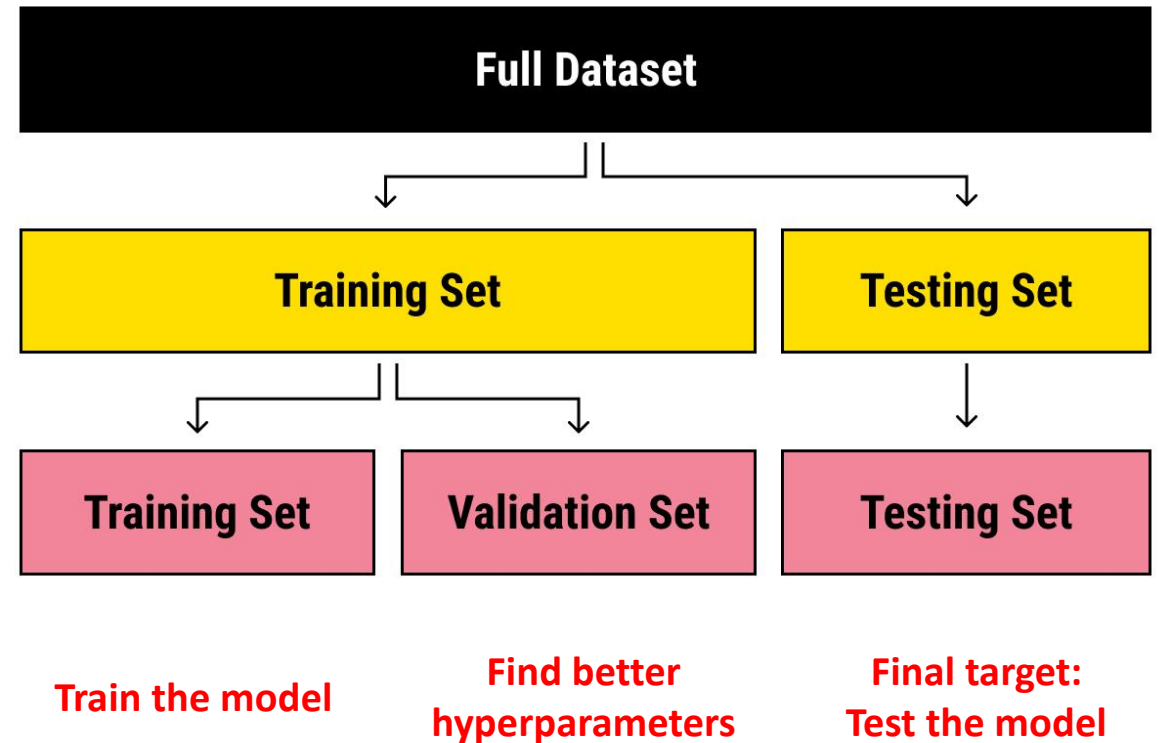


Minus log function

Basics of Deep Learning: Neural Network

✿ Data Preparation

- ✿ Label/Target/Ground truth
- ✿ Split data into Train/Test/Validation sets
- ✿ Use the ratio 3:1 or 4:1 for Train:Validation set
- ✿ Testing set: the data you need to predict.



[1] <https://labeledyourdata.com/>

Basics of Deep Learning: Common NN architectures

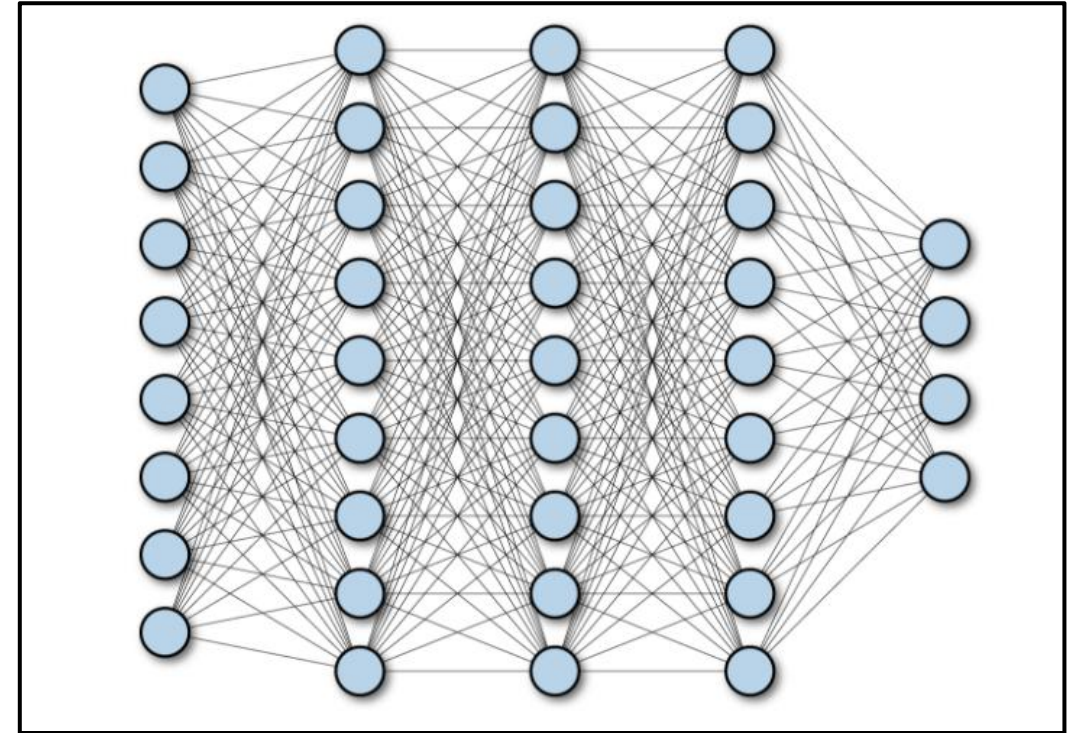
✿ Fully-connected (FC) Layer

- ✧ Node
- ✧ Architecture
- ✧ Optimization

✿ Design Multilayer Perceptron (MLP)

- ✧ Determine the number of input/output dimension
- ✧ Determine the number of middle layers and dimension in each middle layer

Is MLP a black box??

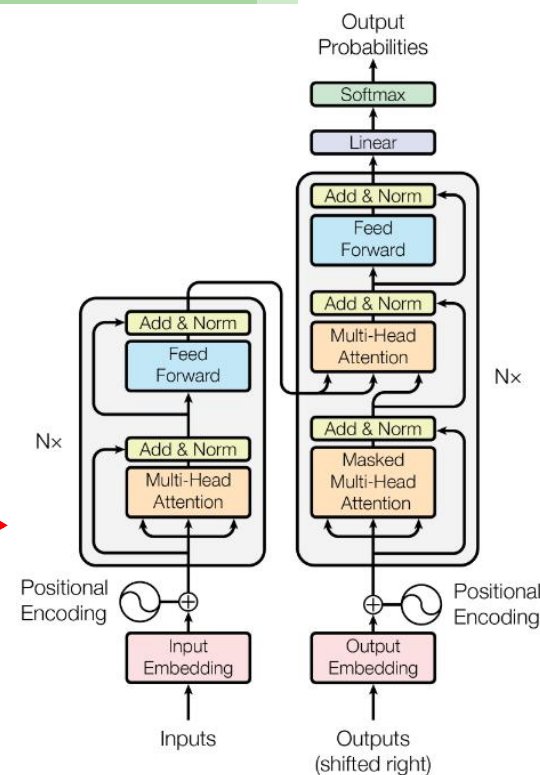
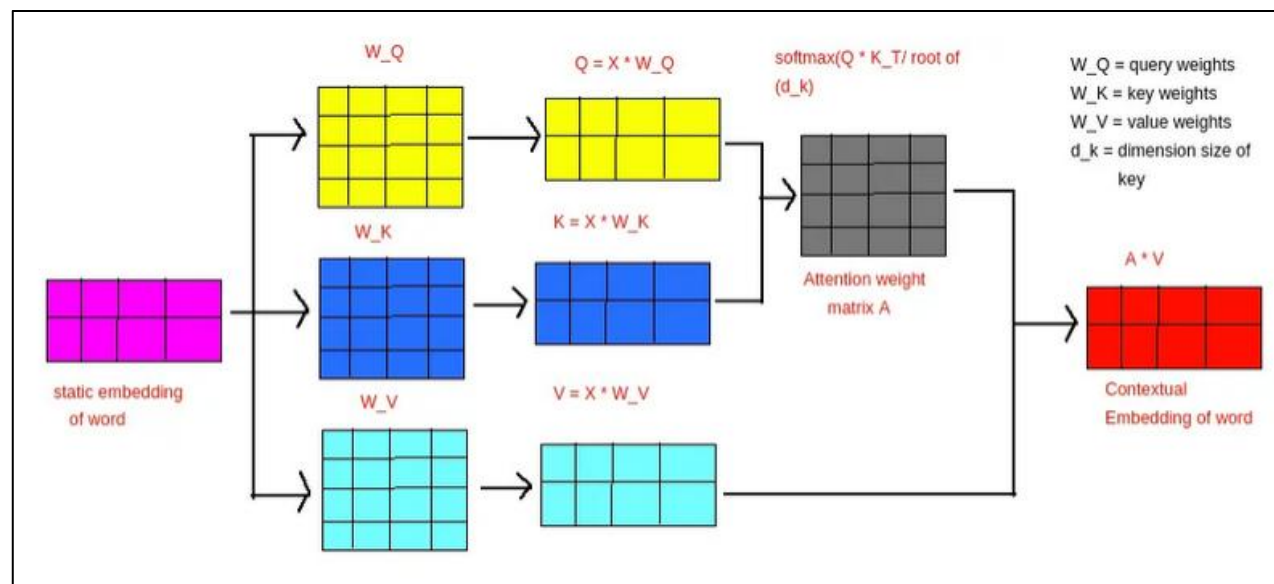


MLP

Basics of Deep Learning: Common NN architectures

🌸 Transformer

🌸 Self-attention module --> Transformer



🌸 Better performance in complex datasets. Most large or foundation models are based on transformer architecture. But, the model is tricky, hard to fine-tune the hyperparameters.

[1] <https://medium.com>

[2] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., 2017. Attention is all you need. NIPS.



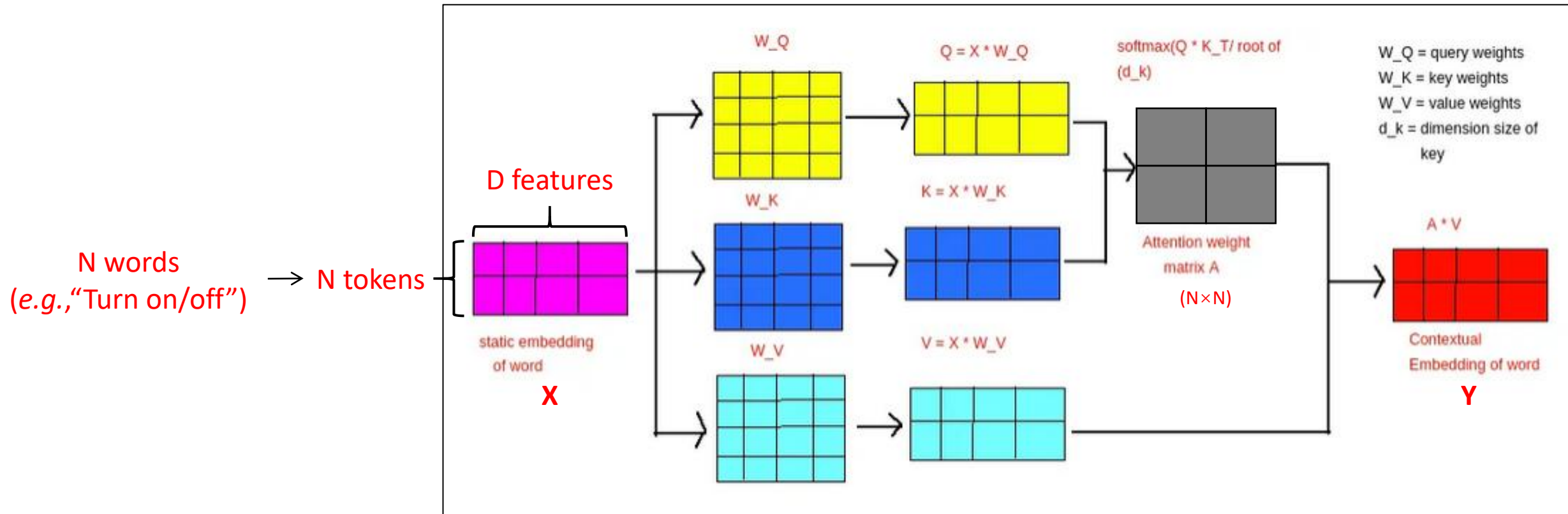
.....

Basics of Deep Learning: Common NN architectures

✿ Transformer: Self-attention Module

✿ **Input X** : embedding of word; **Output Y** : contextual embedding of word

✿ **In Process**: Q, K, V matrices;



Basics of Deep Learning: Common NN architectures

Transformer: Self-attention Module

★ There are 3 essential matrices produced by the module: **Query**, **Key**, **Value**

★ **Q**: request

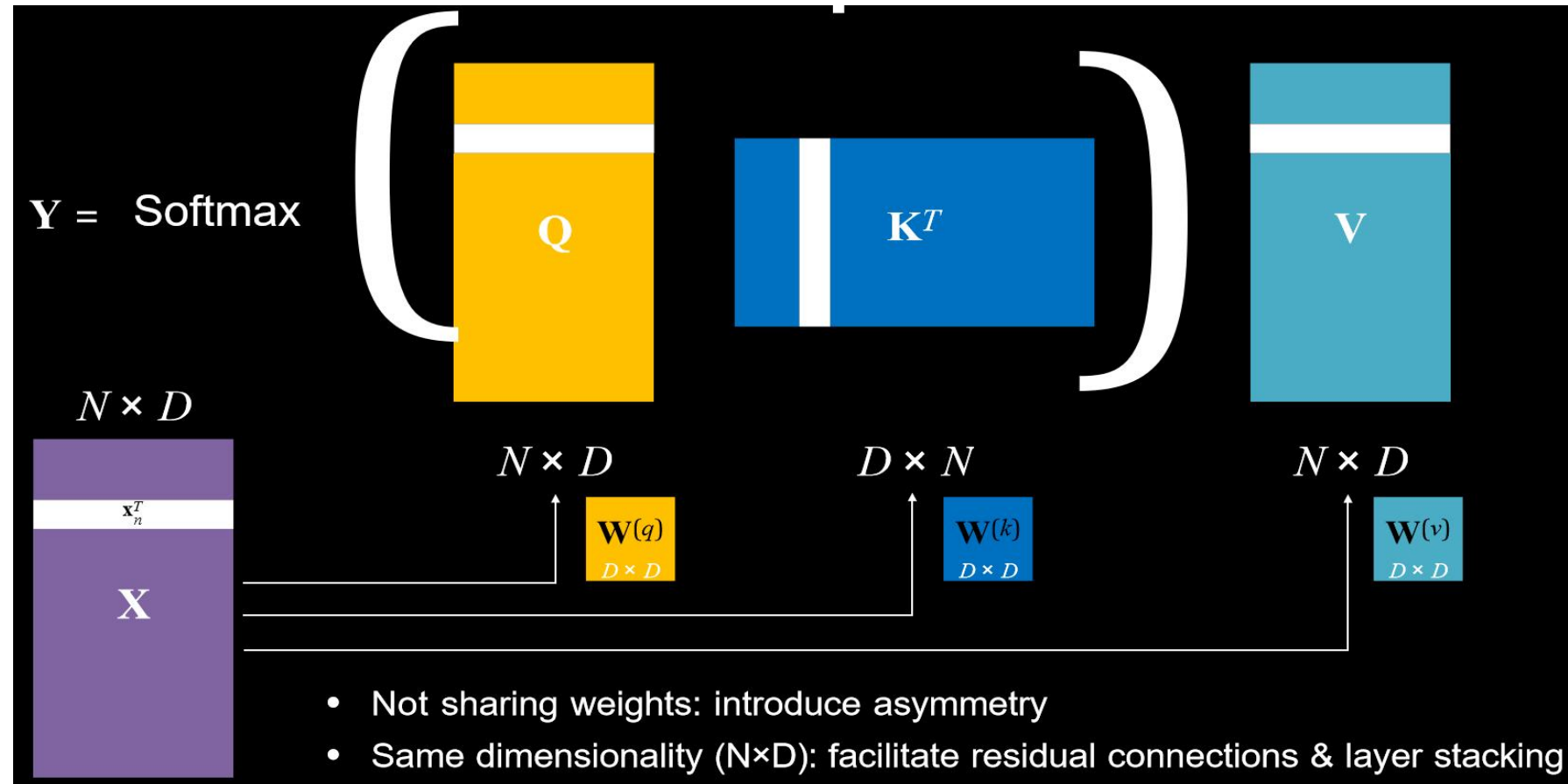
“I like to learn Transformer”

★ **K**: category

“Title of each PPT slide”

★ **V**: content

“PPT content”

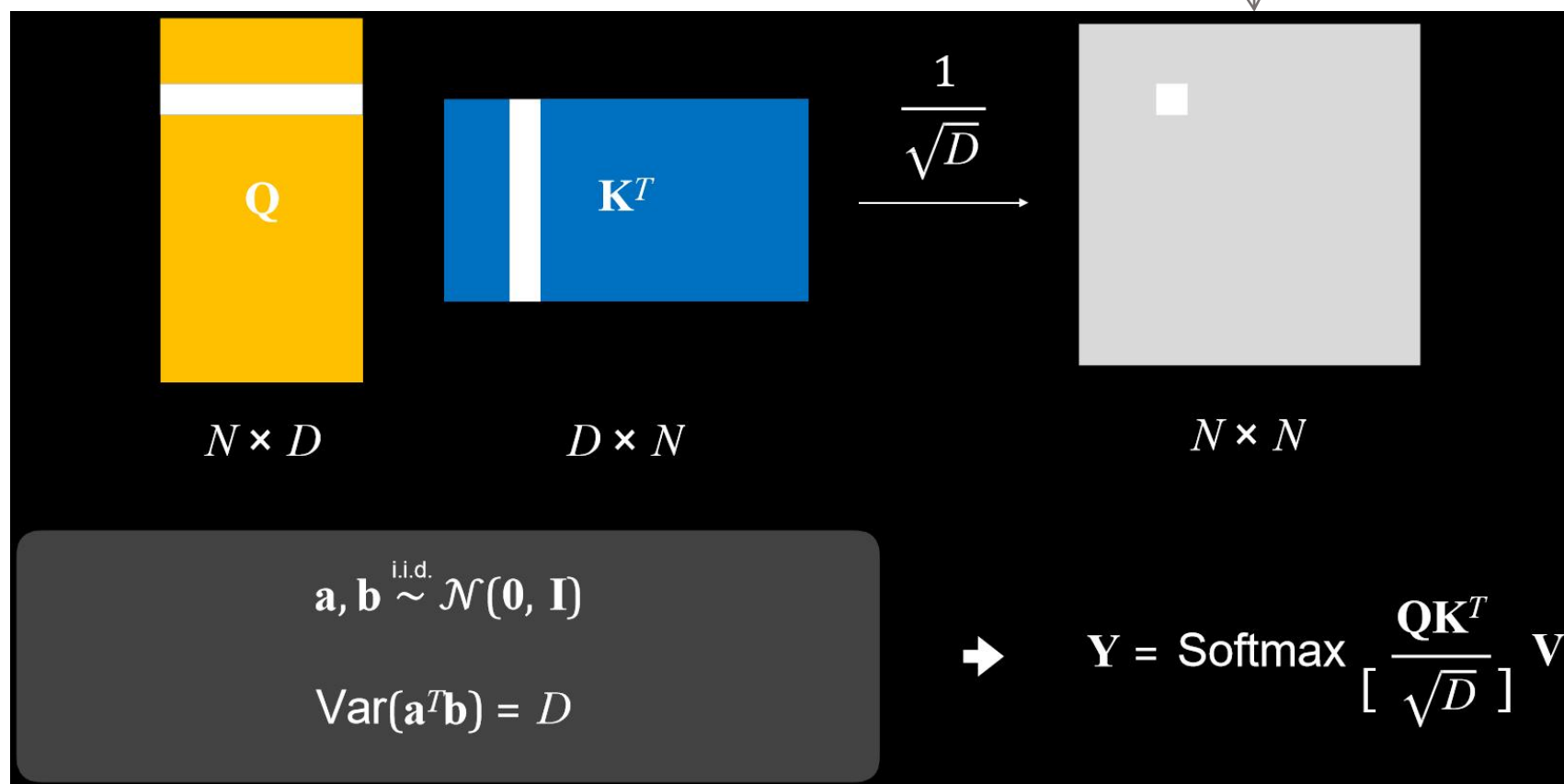


Basics of Deep Learning: Common NN architectures

✿ Transformer: Self-attention Module

✿ Attention matrix **A**

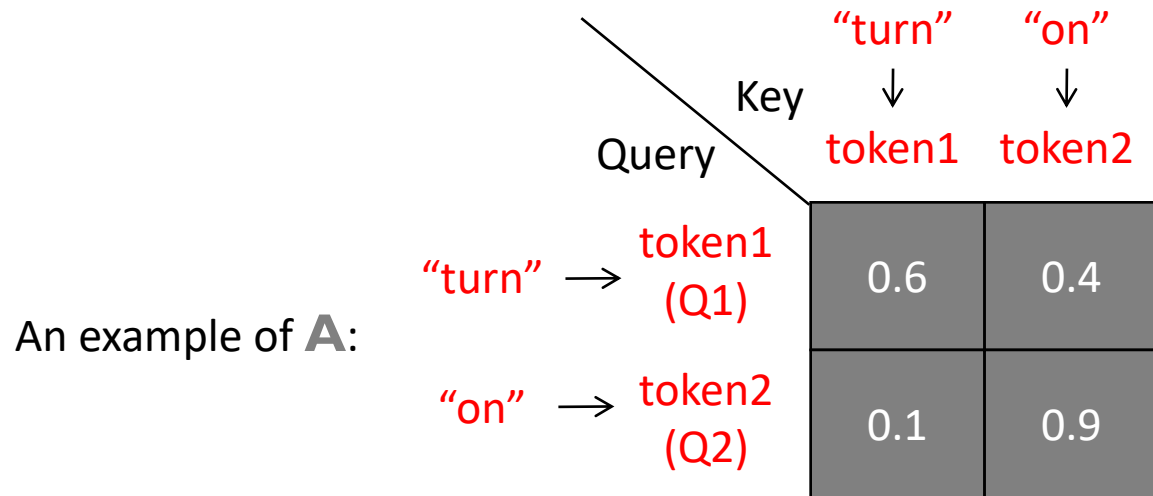
✿ Dot product



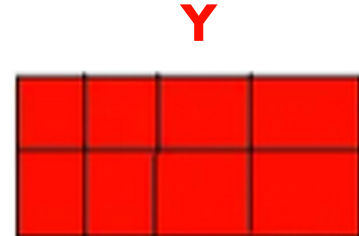
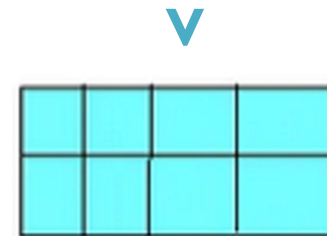
Basics of Deep Learning: Common NN architectures

🌸 Transformer: Self-attention Module

- ✧ Attention matrix $\mathbf{A}$
- ✧ The element of $\mathbf{A}$, a_{ij} , indicates relevance of j th token to i th token.
- ✧ If you use token1 as the Query, token2 is relatively important (0.4).
- ✧ If you use token2 as the Query, token1 is not that important (0.1).



*



Today's PPT



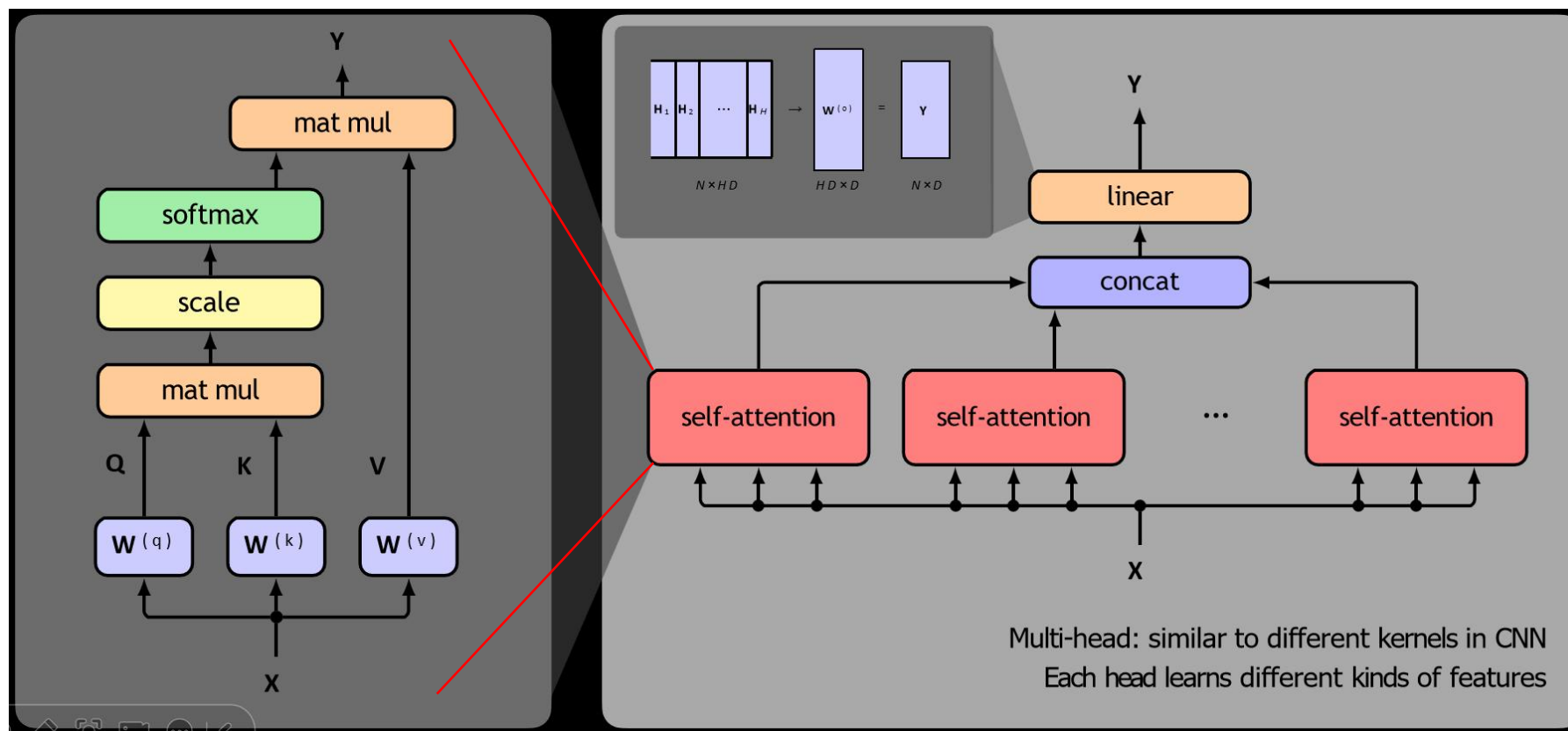
Attended PPT



Basics of Deep Learning: Common NN architectures

✿ Transformer: Multi-head Attention

✿ **Copy** several self-attention modules -> Multi-head attention



Multi-head: similar to different kernels in CNN
Each head learns different kinds of features

✿ Transformer: Structure (Encoder-Decoder)

✿ Self-attention -> Multi-head attention

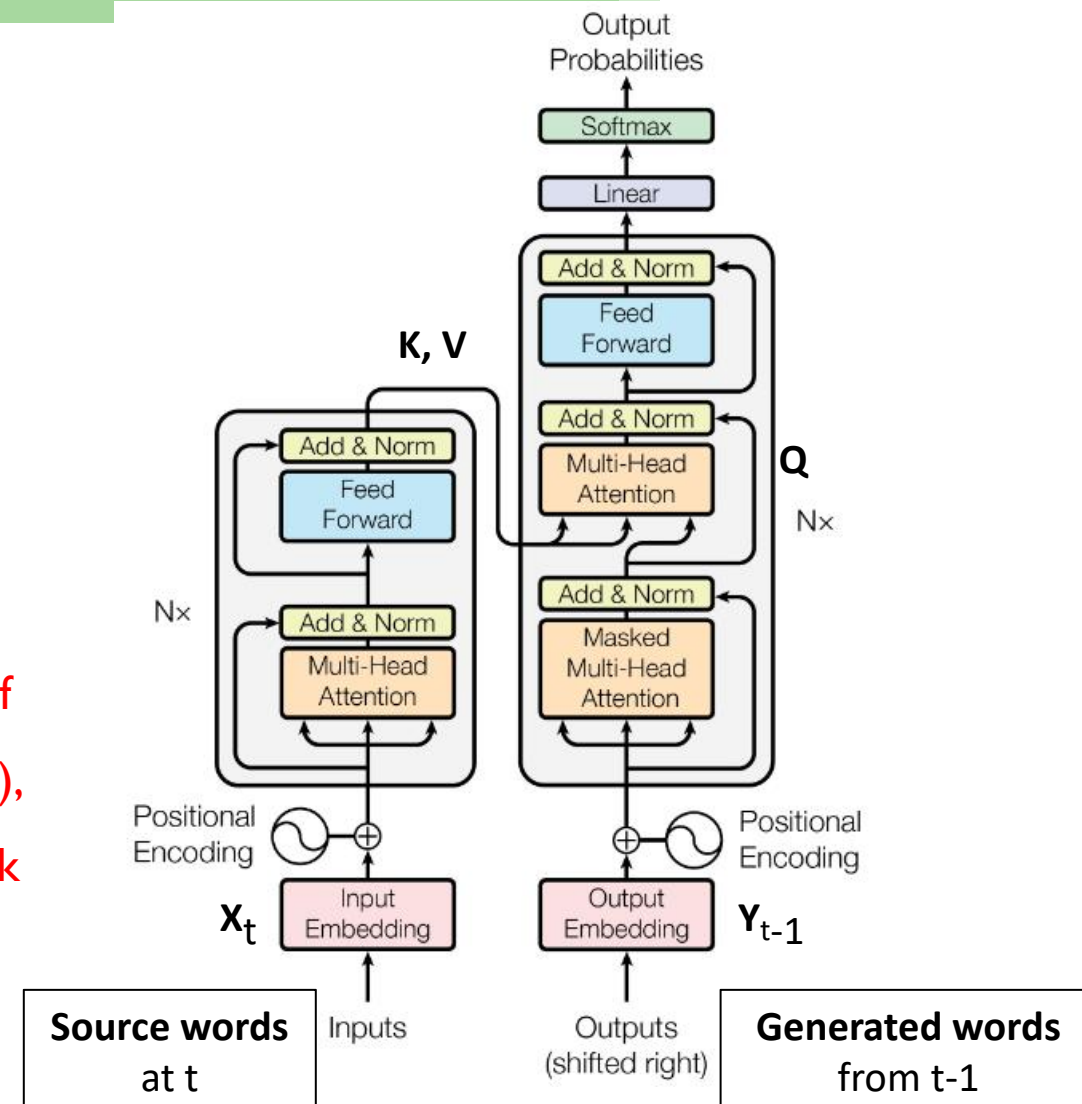
-> Transformer (Encoder-decoder)

✿ Variants of self-attention: Cross-attention, Mamba

Variants of transformer (encoder-decoder):

encoder-only, decoder-only (GPT/Qwen/DeepSeek)

✿ Overall, the transformer (encoder-decoder) outputs a series of words, forming a sentence (e.g., the answer ChatGPT gives), using the source sentence as input (e.g., the question you ask ChatGPT).



Basics of Computer Vision: Background

Classic CV tasks:

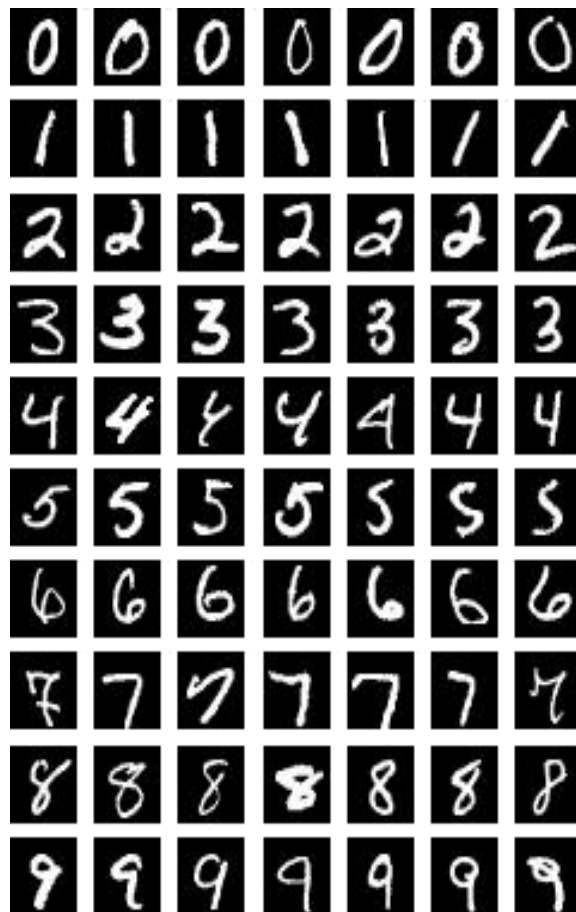
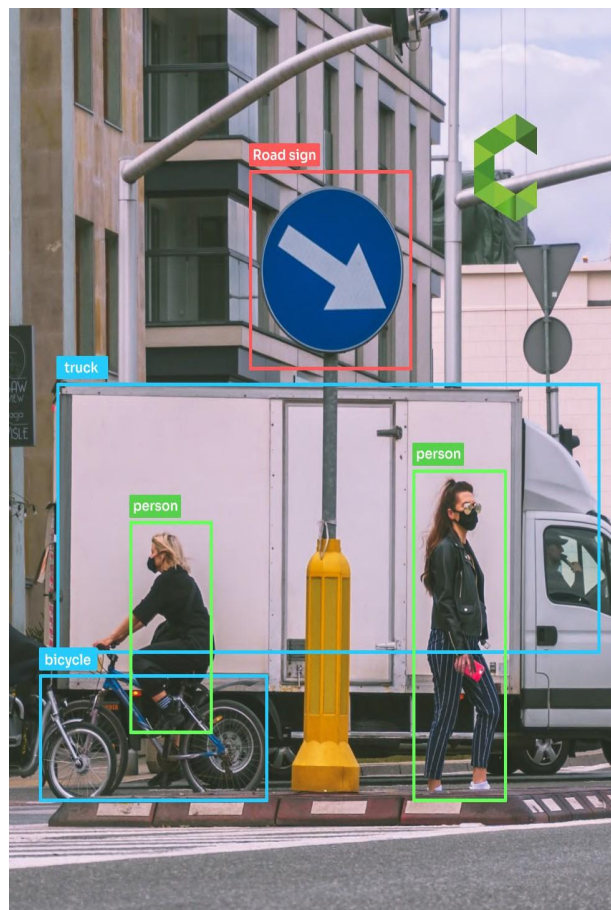


Image Classification



Object Detection



Semantic Segmentation

Basics of Computer Vision: Background

Classic CV tasks:



Image/Video Generation [1]



Image Captioning

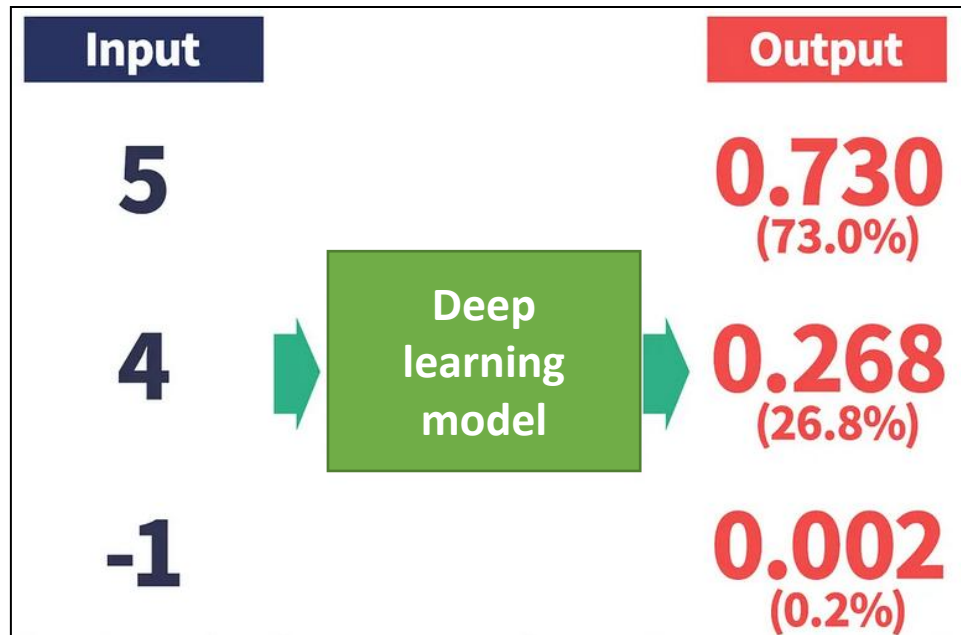
What are these CV tasks for?

[1] <https://www.picsellia.com/post/exploring-stable-diffusion-revolutionizing-image-to-image-generation-in-computer-vision>

Basics of Computer Vision: Background

✿ Why Convolutional Neural Networks?

✿ Dimension: from vector to image



Dimension: 3



Dimension: 2961×3948

[1] <https://medium.com>

Basics of Computer Vision: Convolutional Neural Network

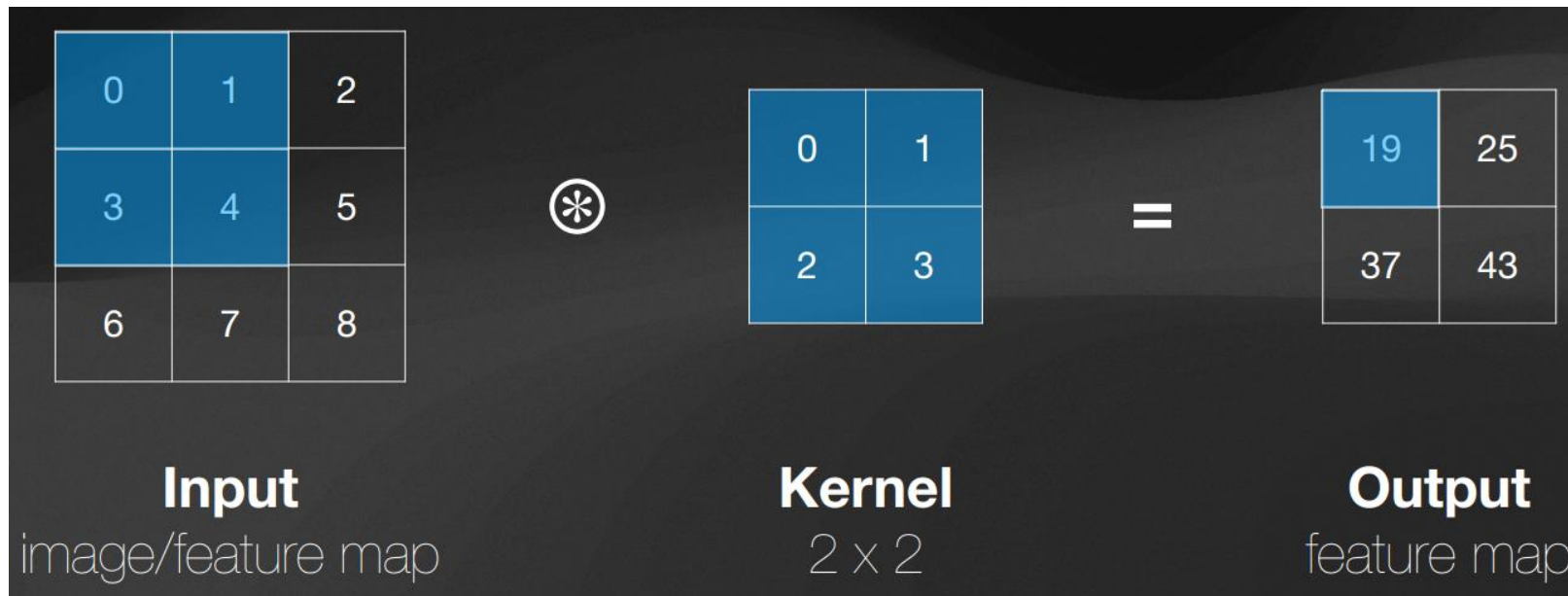
✿ Convolutional Operation (Conv)



- Fewer weights using filters
- Response same to the same image patterns
- Roughly keep spatial information

Basics of Computer Vision: Convolutional Neural Network

✿ Convolutional Operation (Conv)



- Fewer weights using filters
- Response same to the same image patterns
- Roughly keep spatial information

Basics of Computer Vision: Convolutional Neural Network

✿ Convolutional Operation (Conv)

➤ Conv 3×3 filter's convolution

input

2	3	1	4
3	1	3	2
3	0	1	3
0	2	0	1

conv

kernel (3×3)

1	0	-1
1	0	-1
1	0	-1

=

3	-5
2	-3

➤ Conv 3×3 in the model is learnable, f1->f9 are learnable weights.

input

2	3	1	4
3	1	3	2
3	0	1	3
0	2	0	1

conv

kernel (3×3)

f1	f2	f3
f4	f5	f6
f7	f8	f9

=

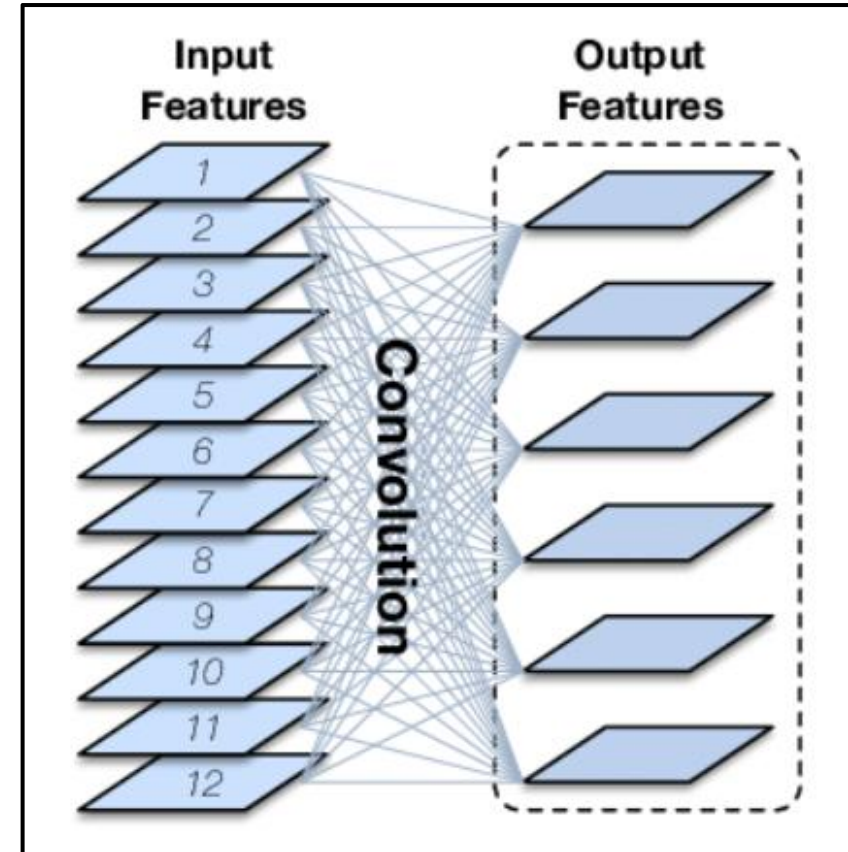
...	...
...	...

[1] <https://8thlight.com/insights/what-is-a-convolution-how-to-teach-machines-to-see-images>

Basics of Computer Vision: Convolutional Neural Network

✿ Conv Layer

- Conv layer
conv and fully-connected operations
 - For example,
conv 3×3 filters, 6
input: $12 \times H \times W$
output: $6 \times h \times w$
- Conv operation:
Each input feature map is processed by 6 filters
- Fully-connected operation:
Each output feature map is a weighted sum of 12 filtered input feature maps

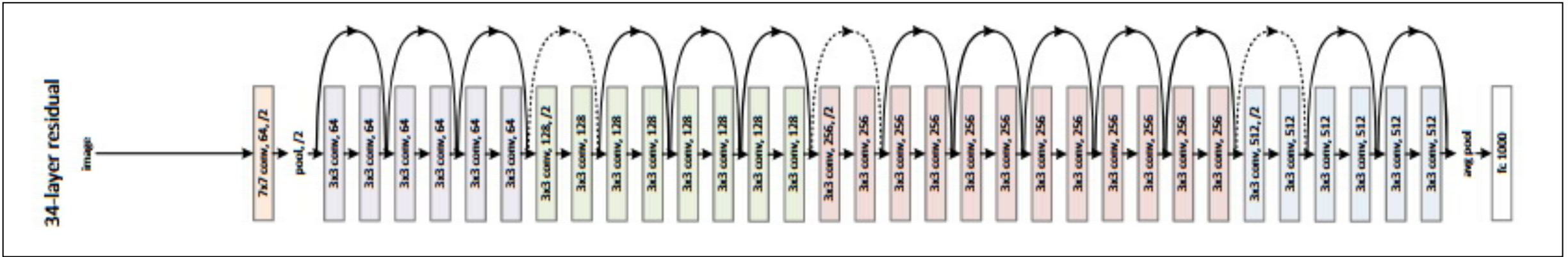


[1] <https://8thlight.com/insights/what-is-a-convolution-how-to-teach-machines-to-see-images>



CNN architecture

- Conv layers are stacked to construct CNN architecture
- Avg/max pooling: get the embedding vector



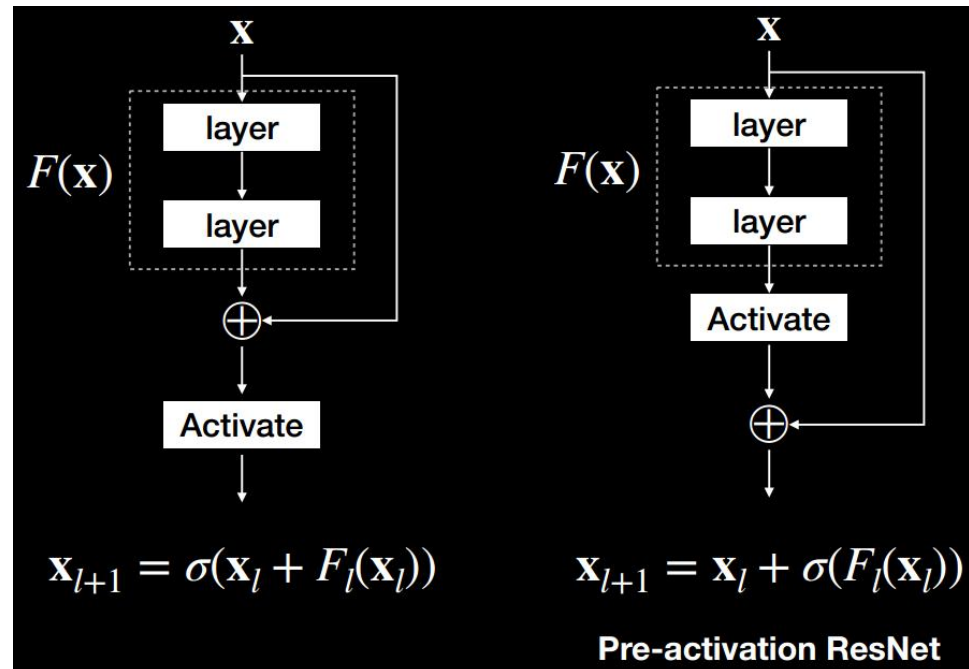
ResNet architecture [1]

[1] He, K., Zhang, X., Ren, S. and Sun, J., 2016. Deep residual learning for image recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 770-778).

Basics of Computer Vision: Common Architectures

✿ ResNet [1]

✧ Skip/shortcut connection



CIVL 7023 Industrialized Construction with Automation and Robotics: A Brief Introduction to Deep Learning and Computer Vision

Thanks!

